SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

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QUESTIONS 10 / 10 completed

Sparse Matrices

SPARSE MATRICES

Represent two sparse matrices (mostly zeros) using arrays of triples [row, col, value]. Read dimensions and non-zero terms for two matrices. Add them, store result in sparse form (sorted
by row then col), and display dense matrix form. Max 10x10; ignore zeros.
Sample Input (Matrix 1):
3 3 3
0 1 5
1 0 8
2 2 2
Sample Input (Matrix 2):
3 3 2
0 1 3
2 2 7
Sample Output (Sum Dense):
0 8 0
8 0 0
0 0 9

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>

#define MAX 10
#define MAX_TERMS 100
typedef struct {
    int r, c, v;
} Triple;

static void sortByRC(Triple a[], int n) {

    for (int i = 0; i < n - 1; i++) {
        for (int j = i + 1; j < n; j++) {
            if (a[j].r < a[i].r || (a[j].r == a[i].r && a[j].c <
                Triple t = a[i];
                a[i] = a[j];
                a[j] = t;
            }
        }
    }
}

```

OUTPUT

```
0 5 0
8 0 0
0 0 2
```

INPUT

```
3 3 3
0 1 5
1 0 8
```

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